

On the water transport behavior and phase transition mechanisms in cold start operation of PEM fuel cell

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HIGHLIGHTS

- Analytical dynamic modeling for cold start of PEM fuel cell is presented.
- Three phase change mechanisms are proposed based on the product water states.
- Transport behavior of the recent reported super-cooled water is performed.
- Membrane water serves as the reasonable state of water production in the model.
- Assumptions on the liquid water and vapor production should be cautiously used.

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ABSTRACT

An analytical dynamic model is proposed to predict the cold start behavior of proton exchange membrane (PEM) fuel cell. Water phase transition mechanisms have been reconstructed based on the five states of water migrating in the fuel cell, composed of water vapor, super-cooled water (liquid water), ice, membrane water (water dissolved in solid electrolyte) and frozen membrane water, in order to explore the reasonable water production and phase transition mechanism. The general water transfer behavior during cold start operation of PEM fuel cell finally evolves into four stages: (1) unsaturated water in ionomer and membrane water production; (2) water in ionomer reaching saturation; (3) over-saturated water in ionomer and quick desorption into liquid and vapor in pores; and (4) significant ice formation. Both non-equilibrium and equilibrium methods to simulate phase transition have been carried out to reveal water transport characteristics in the cold start operation. The assumptions associated with the liquid water and vapor production in the electrochemical reaction extensively involved in the previous modeling studies in the literature should be cautiously used, especially for the dynamic modeling studies. It is worthwhile addressing that the liquid water inside the cathode CL follows the slow increasing and rapid decreasing trend during the cold start operation, which indicates that the liquid water should mostly freeze at the shut-down moment. Furthermore, water prefers to freeze on the interface between catalyst layer (CL) and micro-porous layer (MPL), leading to the liquid water accumulation here migrating towards the membrane side.

1. Introduction

Proton exchange membrane (PEM) fuel cell has been attracting much attention as one of the most promising power plants for portable, stationary, aerospace and military applications due to its low to zero emission, flexible size, high energy conversion efficiency and low operating noise [1]. For the widespread adoption, enough lifetime and reliable operation of PEM fuel cell under a wide range of environments should be realized [2,3]. It is inevitable for PEM fuel cell operating at

subfreezing temperatures. The byproduct water could freeze immediately as the temperature falls down to subfreezing level, which definitely brings about reaction kinetics degradation, porous structure damage and startup failure [4]. Therefore, a solid understanding of cold start behavior is critical to reinforce the comprehensive operational reliability of PEM fuel cell [5].

To figure out the interaction of mass and heat transfer mechanism and the operational characteristics of PEM fuel cell during cold start, experimental measurement and modeling simulation is generally

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Nomenclature

a	water activity; anode
A	geometric area of the fuel cell (m^2)
c	cathode
C	mole concentration (mol m^{-3})
C_p	specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)
d	diameter (m)
D	diffusion coefficient ($\text{m}^2 \text{s}^{-1}$)
EW	equivalent weight of membrane ($1100 \text{ kg kmol}^{-1}$)
F	Faraday's constant ($96487.0 \text{ C mol}^{-1}$)
h	latent heat (J kg^{-1})
I	current density (A m^{-2} , A m^{-3})
k	electrical conductivity (S m^{-1})
K	permeability (m^2)
M	relative molecular weight (kg mol^{-1})
n	moles of electrons production per mole of reactant consumption
n_d	electro-osmotic coefficient ($\text{H}_2\text{O}/\text{OH}^-$)
N	change of moles of gas (mol); flux of the liquid water ($\text{mol m}^{-2} \text{s}^{-1}$)
p	pressure (Pa)
R	ideal gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$); internal resistance of cell (Ωm^2); electrochemical reaction rate (A m^{-3})
s	volume fraction
S	source term ($\text{mol m}^{-3} \text{s}^{-1}$)
ST	stoichiometric ratio
t	time (s)
T	temperature (K)
V	voltage; voltage loss (V)
x, y, z	coordinate position (m)
X	mole fraction

Greek symbols

α	apparent transfer coefficient
δ	thickness (m)
ε	porosity
η	voltage loss (V)
κ	phase change rate (s^{-1})
θ	contact angle ($^\circ$)

λ	water content
μ	dynamic viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)
ρ	density (kg m^{-3}); electrical resistivity (Ωm)
σ	electrical conductivity (S m^{-1})
ω	volume fraction of ionomer in catalyst layer

Superscript and subscripts

0	standard condition
act	activation loss
ave	average
c	capillary pressure; cathode
CL	catalyst layer
eff	effective
e^-	electron
equil	equilibrium
fm	frozen membrane water
g	gas
GDL	gas diffusion layer
H^+	proton
H_2	hydrogen
H_2O	water
i, j	a composition of gas mixture
ice	ice
lq	super-cooled (liquid) water
m	membrane
MPL	micro-porous layer
nf	non-frozen membrane water
nernst	Nernst
ohm	ohmic
out	cell output
O_2	oxygen
BP	bipolar plate
ref	reference condition
sat	saturation state
vap	water vapor
lq-ice	between liquid water and ice
nfm-lq	between non-frozen membrane water and liquid water
nfm-vap	between non-frozen membrane water and vapor
vap-lq	between vapor and liquid water
vap-ice	between vapor and ice

imperative. In the last decade, extensive efforts have been implemented for elucidating the cold start characteristics and startup strategies from various aspects. Three different flow fields were designed by Lin et al. to study their effect on the cold start behavior of PEM fuel cell [6]. They also reported that successful cold start leads to insignificant degradation of cell performance, whereas failed startup gives rise to gradual performance degradation [7]. Segmented fuel cell was employed by them to capture the internal current density and temperature distribution in the PEM fuel cell to better understand the cold start behavior [8]. Various stages of cold start procedure of PEM fuel cell were defined by Gwak et al., including ice growth, ice accumulation with under-saturated vapor, membrane hydration, ice melting and membrane degeneration [9]. Ex-situ scanning electron microscope (SEM) was performed by Tabe et al. to further reveal the ice growth inside the porous electrodes during the cold start tests of PEM fuel cell [10]. A numerical study was carried out by Ko et al. to investigate the effect of catalyst layer (CL) design parameters, including the ionomer fraction and weight ratio of Pt to the carbon support, on the cold start capability of PEM fuel cell [11]. Experimental work was also implemented for comprehensively study the fundamental cold start characteristics correlated with various cell design parameters and control strategy [12,13]. Multi-dimensional cold start models were developed by Meng

[14] and Jiao [15] to elucidate the fundamental physics inside the cell during cold start operation with several key influential parameters, e.g. initial water content in membrane, current density, startup temperature and water vapor content in cathode channel.

Based on the fundamental researches on the cold start behavior of PEM fuel cell, series of startup strategies and optimized cell designs are further developed. An efficient startup scenario was proposed to achieve rapid cold startup of PEM fuel cell through regulating the operating current during the operation [16]. Furthermore, micro-porous layer (MPL) was also used by them to optimize the ice storage ability of the electrode into MPL region for enhancement of cold start performance [17]. A novel method through variable heating and load control was put forward by Zhou et al. to activate the dead individual cells in the cold start operation of fuel cell stacks [18]. Metal foam cathode flow field was also fabricated to replace the conventional parallel channel in PEM fuel cell. Benefit from its extremely porous structure, the resultant cold startup behavior was significantly strengthened [19]. The mechanism of direct catalytic hydrogen-oxygen reaction in anode was stated by Luo et al. to assist warm up the PEM fuel cell under cold start operation [20], which also implies that the cross leakage current caused by the hydrogen crossover through the membrane may be regulated and utilized for heating up the freezing cathode during cold start

operation [21]. Proper humidification strategy is also regarded as an important issue not only under normal operational conditions but also in cold start operation of PEM fuel cell, mainly because of its notable influence on the water transfer characteristics in porous electrodes [22]. As is well known, the deterioration of the cell performance is a common issue after a certain period of operation and cold start cycles, micro-current operation is proposed and validated by Pei et al. to recover the fuel cell performance, which can be also expanded in the performance recovery in the cold start operation [23]. In addition, as the super-cooled water was observed and confirmed in the cold starting fuel cell in recent years, liquid water management strategies are also possibly employed for the reinforcement of the cold start capability [24].

As aforementioned, the cold startup capability of PEM fuel cell has been extensively explored in the literature. However, the water production and phase transition mechanisms are considered to be immature and need to be further consummated. The super-cooled liquid water was confirmed for the first time by Ishikawa et al. through a series of visualization experimental tests on a PEM fuel cell with a transparency cathode [25]. Super-cooled water was observed on the surface of gas diffusion layer (GDL) and freeze at the interface between GDL and CL in cathode [26]. Based on their previous study, the generation mechanism of super-cooled water was preliminarily built and theoretically analyzed [27]. Afterwards, limited investigations were carried out to declare the possible existence of super-cooled water during the cold start of PEM fuel cell, covering the visualization experimental tests and theoretical analysis of the behavior of super-cooled water [9,28,29]. Consequently, the conventional phase transition system which only contains water vapor, ice and membrane water (water dissolved in the ionomer) is not rigorous anymore and possibly unable to accurately predict the water transport inside the PEM fuel cell. Regarding this issue, the phase transition mechanisms inside the PEM fuel cell during startup from subfreezing temperature need to be rebuilt with eager desire.

In this study, dynamic modeling work for the cold start of PEM fuel cell is presented. Three different water production and phase transition mechanisms are proposed based on the possible states of byproduct water, containing water vapor, super-cooled water, ice, non-frozen membrane water and frozen membrane water. The reasonability of different water transition scenarios formulated in the cold start

behavior, as well as the mass and heat transfer processes are discussed for the modeling development. Considering the scarcity of the cold start researches with consideration of super-cooled water, this study qualitatively analyzes the important role of super-cooled water playing in the cold start operation, which should be viewed as an important supplement to the literature. The transport mechanism of super-cooled water during the cold start is further investigated and discussed. The modeling formulation is introduced in the second section. The detailed resultant modeling prediction, proper discussion and suggestions on modeling water transfer behavior during cold start operation have been carried out in the third section, followed by the main conclusions in the last section.

2. Modeling formulation

2.1. Physical problem

Generally, five states of water in the cold start operation of PEM fuel cell should be addressed, composed of water vapor, super-cooled water, ice, non-frozen membrane water and frozen membrane water. Water dissolved in the ionomer of CL and membrane is often defined as membrane water (non-frozen or frozen membrane water), while water behaving in the pore regions of CL, MPL and GDL should be in vapor, super-cooled, and ice states. Basically, the phase of product water in the electrochemical reaction is largely dependent on the site of reaction taking place in cathode CL. Fig. 1 displays the three possible water states generated in the electrochemical reaction, including membrane water, water vapor and liquid water. The specific morphology of the incorporation of carbon supported catalyst (Pt) and ionomer inside CL should dominate the water generation. Reaction taking place in the electrolyte will surely generate membrane water, while water produced in the pores should be in vapor or liquid phase.

In this study, the non-equilibrium water phase transition mechanisms are firstly established in Fig. 2. The different phases of water produced in the reaction determines the beginning point of the phase transition, leading to completely different water transfer characteristics and the resultant cold start behavior. In the following section, the reasonability of these three water generation mechanisms will be carefully analyzed and validated for the modeling application in the literature. Note that in the non-equilibrium phase transition, the

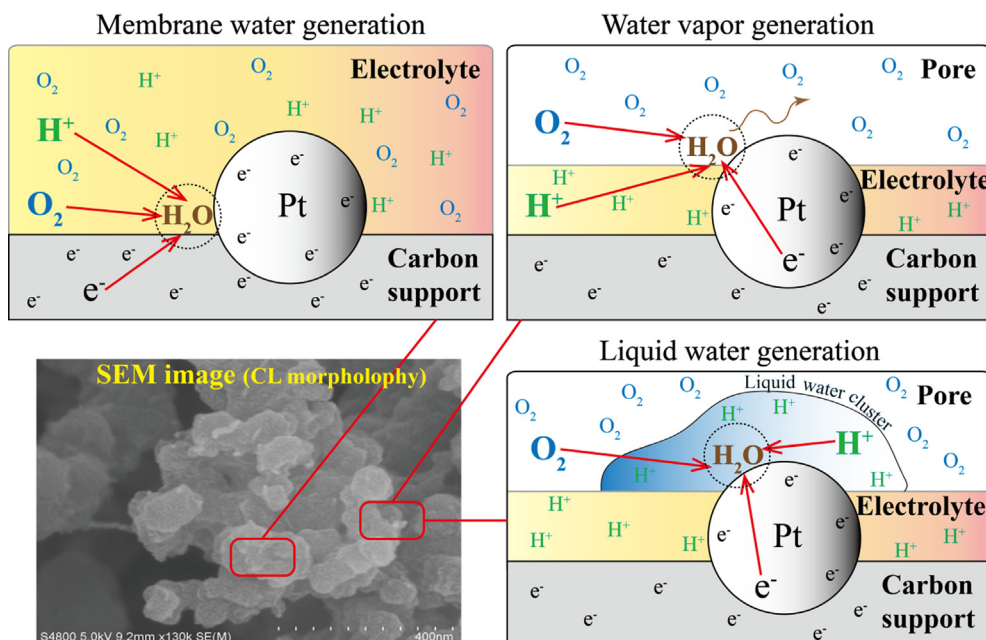


Fig. 1. Schematics of various water generation mechanisms inside cathode CL.

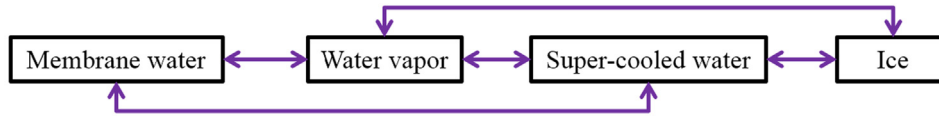
In cathode CL:**In membrane:****In other domains:**

Fig. 2. Non-equilibrium water phase transition mechanisms with consideration of super-cooled water.

membrane water and vapor could reach over-saturated, while for the equilibrium transition the over-saturated state could not be reached, which will be further introduced and discussed in the following sections. As a matter of fact, the phase transition related to the membrane water in this study should be interpreted as the water uptake and release of the ionomer, rather than a genuine kind of water phase change. The resultant startup potential, water and heat variation procedures are compared and discussed. The mass and heat transport are quantified and calculated based on the commercial software package of MATLAB using 1-D matrixes considering all the cell components (bipolar plates, flow channels, GDLs, MPLs, CLs and membrane) perpendicular to the MEA. The physical parameters for the present modeling work are summarized in Table 1. Table 2 gives the initial and operational conditions incorporated in the cold start simulation.

2.2. Assumptions

Several specific assumptions have been included:

- (1) The flow performed in this model is incompressible and laminar because of the weak gas pressure gradient and low Reynolds number.
- (2) The gas flow follows the ideal gas law.
- (3) The membrane is impermeable to the gas flow.
- (4) The effect of gravity is neglected.
- (5) The amount of super-cooled liquid water and vapor is fixed to be zero in the flow channel due to the fast flow rate.

2.3. Cell output

The cell voltage (V_{out} , V) is solved by combining the reversible thermodynamic voltage (V_{nernst} , V), activation overpotential (η_{act} , V) and ohmic overpotential (V_{ohm} , V).

$$V_{\text{out}} = V_{\text{nernst}} + \eta_{\text{act}} + V_{\text{ohm}} \quad (1)$$

The Nernst Equation is widely used to predict the reversible thermodynamic cell voltage [30]:

$$V_{\text{nernst}} = 1.23 - 0.9 \times 10^{-3} \cdot (T - T_0) + \frac{RT}{2F} \ln \left(\frac{p_{\text{H}_2} p_{\text{O}_2}^{0.5}}{p_{\text{H}_2\text{O}}} \right) \quad (2)$$

Here T and T_0 (K) represent the local operating temperature and reference temperature (298 K) at the reaction site. R and F denote the ideal gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$) and Faraday's constant (96487 C mol^{-1}). p_{H_2} , p_{O_2} and $p_{\text{H}_2\text{O}}$ (Pa) refer to the partial pressure for hydrogen, oxygen and water.

In order to distribute the reactant consumption and water

production, the conservations for the protons and electrons transport should be comprehensively implemented. The governing equations for the charge conservation are described as [30]

$$\text{Electrons: } \nabla \cdot (-\sigma_e^{\text{eff}} \nabla V_e) = J_i \quad (3)$$

$$\text{Protons: } \nabla \cdot (-\sigma_H^{\text{eff}} \nabla V_H) = -J_i \quad (4)$$

Here σ_e^{eff} and σ_H^{eff} (S m^{-1}) represent the effective electrical conductivity of the porous electrodes and effective ionic conductivity of polymer electrolyte, respectively [15]. V_e and V_H (V) denotes the electronic and ionic potential in the operating fuel cell, respectively. J_i (A m^{-3}) is the exchange current density (reaction rate of the electrochemical reaction), which is given by the Butler-Volmer Equation [31,32].

$$J_i = J_{\text{ref}} \left(\frac{C_i}{C_{i,\text{ref}}} \right)^{r_i} \left[\exp \left(\frac{\alpha n F}{RT} \eta_{\text{act}} \right) - \exp \left(-\frac{\alpha n F}{RT} \eta_{\text{act}} \right) \right] \quad (5)$$

where α is apparent transfer coefficient (0.5). n is the moles of electrons production (or consumption) per mole of reactant consumption. J_{ref} (A m^{-3}) means the reference current density for the electrochemical reaction. Note that the local current density should be associated with the liquid and ice blockage condition at the reaction site. C_i and C_{ref} (mol m^{-3}) represent the local molar concentration and reference molar concentration of species i (H_2 and O_2) at the reaction site. In order to integrate the influence of super-cooled liquid water and ice on the reaction kinetics, the solution for the exchange current density can be corrected as follows [30]:

$$J_i = (1 - s_{\text{liq}} - s_{\text{ice}}) J_{\text{ref}} \left(\frac{C_i}{C_{i,\text{ref}}} \right)^{r_i} \left[\exp \left(\frac{\alpha n F}{RT} \eta_{\text{act}} \right) - \exp \left(-\frac{\alpha n F}{RT} \eta_{\text{act}} \right) \right] \quad (6)$$

in which s_{liq} and s_{ice} refer to the local volume fraction of liquid water

Table 1
Physical parameters.

Physical parameters	Value
Thickness of BP; Channel; GDL; MPL; CL; membrane	2000; 1000; 270; 30; 10; 30 μm
Active area for reaction	$25 \times 10^{-4} \text{ m}^2$
Equivalent weight of membrane	$1100 \text{ kg kmol}^{-1}$
Volume fraction of ionomer inside CL	40%
Contact angles of GDL; MPL; CL	120°; 150°; 100°
Porosities of GDL; MPL; CL	0.6; 0.4; 0.3
Intrinsic permeability of GDL; MPL; CL	6.2×10^{-12} ; 6.2×10^{-12} ; $6.2 \times 10^{-13} \text{ m}^2$
Electrical conductivities of BP; GDL; MPL; CL	20,000; 300; 300; 300 S m^{-1}
Mass density of membrane	1980 kg m^{-3}

Table 2
Initial and operational parameters of the cold start cases.

Parameters	Value
Feed gases	H ₂ /air
Stoichiometry ratio for anode; cathode	2.0; 2.0
Humidification for anode; cathode inlet	0%
Startup temperature	−20 °C (base condition)
Environmental temperature	−20 °C (base condition)
Inlet pressure	1 atm
Initial water content in the membrane	2.9
Initial liquid water volume fraction inside the cell	0
Initial ice volume fraction inside the cell	0
Initial molar concentration of H ₂ and O ₂	0
Membrane water absorption (and desorption) rate (with vapor)	1 s ^{−1} (base condition)
Membrane water absorption (and desorption) rate (with liquid)	1 s ^{−1} (base condition)
Phase change rate between vapor and liquid (evaporation and condensation)	1 s ^{−1} (base condition)
Phase change rate between vapor and ice (desublimation)	1 s ^{−1} (base condition)
Phase change rate between liquid and ice (fusion and melting)	1 s ^{−1} (base condition)

and ice at the reaction site, respectively. The activation overpotentials for anode and cathode (η_{act}^a and η_{act}^c) could be defined as [30]:

$$\text{Anode: } \eta_{act}^a = -(V_e - V_H +) \quad (7)$$

$$\text{Cathode: } \eta_{act}^c = -(V_H + - V_e -) \quad (8)$$

The ohmic overpotentials contained within the bipolar plates, gas diffusion layer (GDL), micro-porous layer (MPL), catalyst layer (CL) and polymer membrane are accounted:

$$V_{ohm} = -(\Omega_{BP} + \Omega_{GDL} + \Omega_{MPL} + \Omega_{CL} + \Omega_m)I \quad (9)$$

Ω_{BP} , Ω_{GDL} , Ω_{MPL} , Ω_{CL} , Ω_m ($\Omega \cdot m^2$) denote the ohmic resistance of bipolar plates, GDLs, MPLs, CLs and membrane. It should be noted that the Ω_{CL} should consist of the ionic ohmic resistance and electronic ohmic resistance due to the fact that both electrons and hydrogen ion could migrate in the CLs [33].

Table 3
Mass transport properties and correlation.

Parameters	Correlation/value	Unit
Dynamic viscosity of hydrogen	$\mu_{O_2} = 3.205 \times 10^{-3}(T/293.85)^{1.5}(T + 72)^{-1.0}$	kg m ^{−1} s ^{−1}
Dynamic viscosity of oxygen	$\mu_{O_2} = 8.46 \times 10^{-3}(T/292.25)^{1.5}(T + 127)^{-1.0}$	kg m ^{−1} s ^{−1}
Dynamic viscosity of water vapor	$\mu_{vap} = 7.512 \times 10^{-3}(T/291.15)^{1.5}(T + 120)^{-1.0}$	kg m ^{−1} s ^{−1}
Dynamic viscosity of liquid water	$\mu_{liq} = 2.414 \times 10^{-5} \times 10^{247.8/(T-140)}$	kg m ^{−1} s ^{−1}
Hydrogen diffusivity in anode	$D_{H_2} = 1.005 \times 10^{-4}(T/333.15)^{1.5}(101325/P)^{1.0}$	m ² s ^{−1}
Oxygen diffusivity in cathode	$D_{O_2} = 2.652 \times 10^{-5}(T/333.15)^{1.5}(101325/P)^{1.0}$	m ² s ^{−1}
Water vapor diffusivity in anode	$D_{vap} = 1.055 \times 10^{-4}(T/333.15)^{1.5}(101325/P)^{1.0}$	m ² s ^{−1}
Water vapor diffusivity in cathode	$D_{vap} = 2.982 \times 10^{-5}(T/333.15)^{1.5}(101325/P)^{1.0}$	m ² s ^{−1}
Effective diffusivity of the gas i (O ₂ and vapor) in the electrode j (GDL, MPL and CL)	$D_i^{eff} = D_i \epsilon_j^{1.5} (1 - s_{liq} - s_{ice})^{1.5}$	m ² s ^{−1}
Mass density of liquid water and ice	$\rho_{liq} = 990, \rho_{ice} = 920$	kg m ^{−3}
Saturated vapor pressure	$\log_{10}(\frac{P_{sat}}{101325}) = -2.1794 + 0.02953(T - 273.15) - 9.1837 \times 10^5(T - 273.15)^2 + 1.4454 \times 10^{-7}(T - 273.15)^3$	Pa
Liquid water surface tension coefficient	$\sigma_{liq} = -0.0001676 \times T + 0.1218$	N m ^{−1}
Equilibrium water content	$\lambda_{equil} = \begin{cases} 0.043 + 17.81a - 39.85a^2 + 36.0a^3 & \text{if } 0 \leq a < 1 \\ 14.0 + 1.4(a - 1) & \text{if } 1 < a \leq 3 \end{cases}$	
Reaction transfer coefficients	$\alpha = 0.5$	

2.4. Mass and heat transport

2.4.1. Water transport and phase transition

Several mass conservation equations should be accounted based on the five states of water. The mass conservation equation for non-frozen membrane water is solved by

$$\frac{\rho_m}{EW} \cdot \frac{\partial(\omega \lambda_{nfm})}{\partial t} = \frac{\rho_m}{EW} \cdot \frac{\partial^2(\omega^{1.5} D_{nfm} \lambda_{nfm})}{\partial x^2} + S_{nfm} \quad (10)$$

ρ_m (kg m^{−3}) is the mass density of membrane. EW (kg mol^{−1}) represents the equivalent weight of membrane. ω is defined as the volume fraction of ionomer in CL. λ_{nfm} is the non-frozen membrane water content. t (s) is time. S_{nfm} (mol m^{−3} s^{−1}) denotes the source term for the non-frozen membrane water. D_{nfm} (m² s^{−1}) is the diffusivity of membrane water inside the ionomer and can be presented by a piecewise function [34]:

$$D_{nfm} = \begin{cases} 2.69266 \times 10^{-10} & \text{if } \lambda_{nfm} \leq 2 \\ 10^{-10} \cdot \exp\left(2416\left(\frac{1}{303} - \frac{1}{T}\right)\right) \cdot (0.87(3 - \lambda_{nfm}) + 2.95(\lambda_{nfm} - 2)) & \text{if } 2 < \lambda_{nfm} \leq 3 \\ 10^{-10} \cdot \exp\left(2416\left(\frac{1}{303} - \frac{1}{T}\right)\right) \cdot (2.9514(4 - \lambda_{nfm}) + 1.642454(\lambda_{nfm} - 3)) & \text{if } 3 < \lambda_{nfm} \leq 4 \\ 10^{-10} \cdot \exp\left(2416\left(\frac{1}{303} - \frac{1}{T}\right)\right) \cdot (2.563 - 0.33\lambda_{nfm} + 0.0264\lambda_{nfm}^2 - 0.000671\lambda_{nfm}^3) & \text{if } \lambda_{nfm} > 4 \end{cases} \quad (11)$$

The diffusivity relies heavily on the membrane hydration. This correlation is widely used in the modeling formulation [18,20,33]. In this study, the water dissolved in the ionomer is defined as membrane water, which is solved based on the molar concentration of water maintained in the membrane (c_{nfm} , mol m^{−3}).

$$\lambda_{nfm} = \frac{EW}{\rho_m} c_{nfm} \quad (12)$$

A threshold value for the maximum water uptake potential, saturated membrane water content (λ_{sat}), is empirically given by [15] derived from the experimental tests carried out by Thompson et al. [35].

$$\lambda_{sat} = \begin{cases} 4.837 \text{ if } T < 223.15 \text{ K} \\ [-1.304 + 0.01479T - 3.594 \times 10^{-5}T^2]^{-1} \text{ if } 223.15 \text{ K} \leq T < T_{freeze} \\ > \lambda_{nfm} \text{ if } T \geq T_{freeze} \end{cases} \quad (13)$$

The saturated membrane water content is often used to determine the water desorption and absorption of the electrolyte in the fuel cell operation [30]. In the membrane, water can also freeze into frozen membrane water at subfreezing temperature. The mass conservation can be calculated through

$$\frac{\rho_m}{EW} \cdot \frac{\partial(\omega \lambda_{fm})}{\partial t} = S_{fm} \quad (14)$$

in which λ_{fm} and S_{fm} ($\text{mol m}^{-3} \text{s}^{-1}$) represent the water content and source term for frozen membrane water, respectively.

With regard to the super-cooled liquid water and vapor, their mass conservation can be solved by

$$\frac{\partial(\varepsilon \rho_{lq} s_{lq})}{\partial t} = \frac{\partial^2 \left(\rho_{lq} \frac{K_{lq}}{\mu_{lq}} p_{lq} \right)}{\partial x^2} + S_{lq} \quad (15)$$

and

$$\frac{\partial(\varepsilon(1 - s_{lq} - s_{ice}) \rho_g C_{vap})}{\partial t} = \frac{\partial^2 (\rho_g D_{vap}^{eff} C_{vap})}{\partial x^2} + S_{vap} \quad (16)$$

The convection term is neglected in this study due to the fact that the mass convection is more severe at the interface of GDL and flow channel. ε is the porosity of the porous electrode. ρ_{lq} (kg m^{-3}), K_{lq} (m^2), μ_{lq} ($\text{kg m}^{-1} \text{s}^{-1}$), s_{lq} and S_{lq} ($\text{kg m}^{-3} \text{s}^{-1}$) serve as the mass density, permeability, dynamic viscosity, local volume fraction and source term of liquid water, respectively. Note that liquid pressure (p_{lq} , Pa) is performed in the liquid conservation instead of liquid saturation in order to maintain the continuity of capillary pressure and achieve liquid saturation jump at the adjacent porous layers with different hydrophobicity, permeability and porosity. ρ_g (kg m^{-3}) means the mass density of gas mixture. D_{vap}^{eff} ($\text{m}^2 \text{s}^{-1}$), C_{vap} (mol m^{-3}) and S_{vap} ($\text{kg m}^{-3} \text{s}^{-1}$) are the effective diffusivity, molar concentration and source term for water vapor. The detailed transport properties of gas species and liquid are provided in Table 3.

As for the ice, the mass conservation is

$$\frac{\partial(\varepsilon s_{ice} \rho_{ice})}{\partial t} = S_{ice} \quad (17)$$

The mass transport of super-cooled water, water vapor and ice should be solved in flow channels, GDLs, MPLs and CLs. s_{ice} is the volume fraction of ice. ρ_{ice} (kg m^{-3}) and S_{ice} ($\text{kg m}^{-3} \text{s}^{-1}$) are the mass

density and source term for ice. Through the analysis of mass conservation equation, the transport behavior of different water states can be obtained.

In this work, the super-cooled water is also carefully considered to develop several more thorough phase change mechanisms for PEM fuel cell during the cold start operation. The source terms for membrane water (S_{nfm}), frozen membrane water (S_{fm}), super-cooled water (S_{lq}), water vapor (S_{vap}) and ice (S_{ice}), can be summarized as:

$$S_{nfm} = \begin{cases} -S_{nfm-fm} \text{ (solved in membrane)} \\ -S_{nfm-vap} - S_{nfm-lq} + S_{EOD} + S_{Prod} \text{ (solved in cathode CL)} \\ -S_{nfm-vap} - S_{nfm-lq} - S_{EOD} \text{ (solved in anode CL)} \end{cases} \quad (18)$$

$$S_{fm} = S_{nfm-fm} \text{ (solved in membrane)} \quad (19)$$

$$S_{lq} = \begin{cases} S_{vap-lq} - S_{lq-ice} + S_{nfm-lq} M_{H_2O} \text{ (solved in CLs)} \\ S_{vap-lq} - S_{lq-ice} \text{ (solved in MPLs, GDLs)} \end{cases} \quad (20)$$

$$S_{vap} = \begin{cases} -S_{vap-lq} - S_{vap-ice} + S_{nfm-vap} M_{H_2O} \text{ (solved in CLs)} \\ -S_{vap-lq} - S_{vap-ice} \text{ (solved in MPLs, GDLs)} \end{cases} \quad (21)$$

$$S_{ice} = S_{vap-ice} + S_{lq-ice} \text{ (solved in CLs, MPLs and GDLs)} \quad (22)$$

where S_{i-j} means the phase change source term from water species i to j . The detailed source terms for water phase transition and transfer are summarized in Table 4. Note that the water molecular is always combined with the hydrogen ion to immigrate through the membrane and the ionomer in CLs from anode towards cathode, known as electro-osmosis drag effect, thus the source terms (S_{EOD} , $\text{mol m}^{-3} \text{s}^{-1}$) being reasonably given for the non-frozen membrane water firstly in this study, as solved in Eq. (18).

$$S_{EOD} = n_d \frac{J_i}{nF} \quad (23)$$

Accordingly, the electro-osmotic drag coefficient (n_d , number of water molecules dragged per hydrogen ion through the electrolyte) is a common parameter for the water transport in the membrane, which is highly dependence on the water uptake of the polymer membrane [15]:

$$n_d = \frac{2.5 \lambda_{nfm}}{22} \quad (24)$$

It is generally formulated to solve the electro-osmosis drag process in the polymer membrane [5,30,36]. This S_{Prod} ($\text{mol m}^{-3} \text{s}^{-1}$) represents the source term for the water generation in the cathode semi-reaction. Note that S_{Prod} solved for the water generation should be based on the generation of water state, including liquid, vapor and

Table 4
Source terms for water phase transition and water mass transfer.

Water species	Phase change function	Unit
Vapor – liquid water (evaporation and condensation)	$S_{vap-lq} = \begin{cases} \kappa_{vap-lq}(1 - s_{lq} - s_{ice}) \frac{(C_{vap} - C_{sat})}{RT} M_{H_2O} \text{ if } C_{vap} \geq C_{sat} \\ \kappa_{vap-lq}(s_{lq} + s_{ice}) \frac{(C_{vap} - C_{sat})}{RT} M_{H_2O} \text{ if } C_{vap} < C_{sat} \end{cases}$	$\text{kg m}^{-3} \text{s}^{-1}$
Vapor – ice (desublimation)	$S_{vap-ice} = \begin{cases} \kappa_{vap-ice}(1 - s_{lq} - s_{ice}) \frac{(C_{vap} - C_{sat})}{RT} M_{H_2O} \text{ if } C_{vap} \geq C_{sat} \\ 0 \text{ if } C_{vap} < C_{sat} \end{cases}$	$\text{kg m}^{-3} \text{s}^{-1}$
Liquid water – ice (freezing and melting)	$S_{lq-ice} = \begin{cases} \kappa_{lq-ice} \varepsilon s_{lq} \rho_{lq} \text{ if } T < T_{freeze} \\ -\kappa_{lq-ice} \varepsilon s_{ice} \rho_{ice} \text{ if } T \geq T_{freeze} \end{cases}$	$\text{kg m}^{-3} \text{s}^{-1}$
Non-frozen membrane water – vapor (mass transfer)	$S_{nfm-vap} = \kappa_{nfm-vap} \frac{\rho_m}{EW} (\lambda_{nfm} - \lambda_{equil})(1 - s_{lq} - s_{ice})$	$\text{kmol m}^{-3} \text{s}^{-1}$
Non-frozen membrane water – super-cooled water (mass transfer)	$S_{nfm-lq} = \begin{cases} \kappa_{nfm-lq} \frac{\rho_m}{EW} (\lambda_{nfm} - \lambda_{sat})(1 - s_{lq} - s_{ice}) \text{ if } \lambda_{nfm} \geq \lambda_{sat} \\ 0 \text{ if } \lambda_{nfm} < \lambda_{sat} \end{cases}$	$\text{kmol m}^{-3} \text{s}^{-1}$
Non-frozen membrane water – frozen membrane water (mass transfer)	$S_{nfm-fm} = \begin{cases} \kappa_{nfm-fm} \frac{\rho_m}{EW} (\lambda_{nfm} - \lambda_{sat}) \text{ if } \lambda_{nfm} \geq \lambda_{sat} \\ \kappa_{nfm-fm} \frac{\rho_m}{EW} \lambda_{fm} \text{ if } \lambda_{nfm} < \lambda_{sat} \end{cases}$	$\text{kmol m}^{-3} \text{s}^{-1}$

membrane water. It is temporally given to the source term of membrane water here, which will be further discussed in the following section.

$$S_{\text{Prod}} = \frac{J_i}{nF} \quad (25)$$

In this study, continuous pressures of gas and liquid are all taken into consideration, which matches the trend in the literature in order to achieve the liquid water saturation jump at the interface of the adjacent porous layers, and the capillary pressure is given by [37,38]:

$$P_c = P_g - P_{lq} \quad (26)$$

and

$$P_c = \begin{cases} \sigma_{lq} \cos \theta \left(\frac{\varepsilon}{K_0} \right)^{0.5} [1.42(1 - s_{lq}) - 2.12(1 - s_{lq})^2 + 1.26(1 - s_{lq})^3] & \text{if } \theta < 90^\circ \\ \sigma_{lq} \cos \theta \left(\frac{\varepsilon}{K_0} \right)^{0.5} [1.42s_{lq} - 2.12s_{lq}^2 + 1.26s_{lq}^3] & \text{if } \theta > 90^\circ \end{cases} \quad (27)$$

This Equation is provided by the well-known Leverett function, which is widely applied in solving the liquid water transfer in the porous media driven by the capillary pressure [30,39,40]. σ_{lq} (N m^{-1}) represents the surface tension of liquid water. θ ($^\circ$), ε and K_0 (m^2) signify the contact angle, porosity and intrinsic permeability of the local electrode. The freezing point depression has been also taken into account depending on the surface tension of water at 273.15 K ($\sigma_{273.15}$, N m^{-1}), hydrophobicity (θ , $^\circ$), mass density of ice (ρ_{ice} , kg m^{-3}), latent heat of fusion (h_{fushn} , J kg^{-1}) and the pore radius of the local porous media (r , m) [15]:

$$\Delta T = \frac{T_{\text{normal}} \sigma_{273.15} \cos \theta}{\rho_{\text{ice}} h_{\text{fushn}} r} \quad (28)$$

Therefore, the actual freezing temperature should be solved below

$$T_{\text{freeze}} = 273.15 - \Delta T \quad (29)$$

2.4.2. Reactant gases transport

The mass conservations for hydrogen and oxygen can be expressed by:

$$\frac{\partial(\varepsilon(1 - s_{lq} - s_{\text{ice}})\rho_g C_i)}{\partial t} = \frac{\partial^2(\rho_g D_i^{\text{eff}} C_i)}{\partial x^2} + S_i \quad (30)$$

ρ_g (kg m^{-3}) means the mass density of the gas mixture. D_i^{eff} ($\text{m}^2 \text{s}^{-1}$) and C_i (mol m^{-3}) are the effective diffusivity and molar concentration of gas species i ($i = \text{H}_2$ and O_2). The source term for species i (S_i , $\text{kg m}^{-3} \text{s}^{-1}$) is dependent on the consumption in the electrochemical reaction. The detailed transport properties of gas species are summarized in Table 3.

2.4.3. Heat transport

The energy conservation equation is solved by

$$\frac{\partial}{\partial t}((\rho C_p)_{\text{fl,sl}}^{\text{eff}} T) = \frac{\partial^2(k_{\text{fl,sl}}^{\text{eff}} T)}{\partial x^2} + S_T \quad (31)$$

$(\rho C_p)_{\text{fl,sl}}^{\text{eff}}$ and $k_{\text{fl,sl}}^{\text{eff}}$ represent the effective volumetric heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$) and effective thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$) of the local electrode. S_T (W m^{-3}) is the source term for heat, including the reversible heat ($S_{T,\text{rev}}$, W m^{-3}), activation heat ($S_{T,\text{act}}$, W m^{-3}) and ohmic heat ($S_{T,\text{ohm}}$, W m^{-3}) from the electrochemical reaction and latent heat ($S_{T,\text{lant}}$, W m^{-3}) from the water phase change.

$$S_T = \begin{cases} S_{T,\text{act}} + S_{T,\text{ohm}} + S_{T,\text{lant}} & \text{(solved in anode CL)} \\ S_{T,\text{ohm}} + S_{T,\text{lant}} & \text{(solved in membrane)} \\ S_{T,\text{rev}} + S_{T,\text{act}} + S_{T,\text{ohm}} + S_{T,\text{lant}} & \text{(solved in cathode CL)} \\ S_{T,\text{ohm}} + S_{T,\text{lant}} & \text{(solved in GDLs, MPLs and CLs)} \\ S_{T,\text{ohm}} & \text{(solved in BPs)} \end{cases} \quad (32)$$

Considering the friendly reaction kinetics of anode reaction, the reversible heat is only solved in cathode CL in this study. The various heat source terms can be expressed as:

$$S_{T,\text{rev}} = -\frac{\Delta S T}{nF} \cdot J_i \quad (33)$$

$$S_{T,\text{act}} = \eta_{\text{act}} \cdot J_i \quad (34)$$

$$S_{T,\text{ohm}} = \begin{cases} \|\nabla V_e\|^2 \sigma_e^{\text{eff}} + \|\nabla V_{H^+}\|^2 \sigma_{H^+}^{\text{eff}} & \text{(solved in CLs)} \\ \|\nabla V_e\|^2 \sigma_e^{\text{eff}} & \text{(solved in other domains)} \end{cases} \quad (35)$$

$$S_{T,\text{lant}} = S_{i-j} \cdot h_{i-j} \quad (36)$$

in which ΔS is the change in entropy ($\text{J mol}^{-1} \text{K}^{-1}$) with 1 mol H_2 consumed. S_{i-j} ($\text{kmol m}^{-3} \text{s}^{-1}$, $\text{kg m}^{-3} \text{s}^{-1}$) and h_{i-j} (J kg^{-1}) are the source term and latent heat for phase transition between water species i and j , respectively. As previous stated, both the ohmic heat of electron and hydrogen ion transfer should be included in the ohmic heat calculation. The thermal properties are given in Table 5.

Basically, this model is analytically solved based on the commercial software package of MATLAB. Series of 1D matrixes are built and used to describe all the mechanisms inherent in the PEM fuel cell, such as the mass and heat transport (species and temperature distribution), electrochemical reactions in CLs and boundary conditions used in the calculation. For better understand the algorithm how the group of equations is analytically solved, Fig. 3 gives the flow chart of the modeling calculation.

3. Results and discussion

The mass and thermal transport in PEM fuel cell relies heavily upon the water generation, removal and freezing process during the PEM fuel cell operation at subfreezing temperature. In most of the previous modeling works in the literature, the water phase transition process is often oversimplified and has not received deserved consideration, possibly providing unreasonable water transport behavior inside the porous electrodes. Generally, the state of product water always depends

Table 5
Thermal properties.

Parameters	Correlation/value	Unit
Specific heat capability of membrane; CL; MPL; GDL; BP	$C_p = 833; 3300; 2000; 568; 1580$	$\text{J kg}^{-1} \text{K}^{-1}$
Heat transfer coefficient between bipolar plates and surrounding	$H = 100$	$\text{W m}^{-2} \text{K}^{-1}$
Intrinsic thermal conductivities of membrane; CL; MPL; GDL; BP	$k = 0.95; 1.0; 1.0; 20$	$\text{W m}^{-1} \text{K}^{-1}$
Latent heat for phase change between super-cooled water and vapor (evaporation and condensation)	$h_{\text{lq-vap}} = 2.308 \times 10^{-6}$	J kg^{-1}
Latent heat for phase change between vapor and ice (desublimation)	$h_{\text{vap-ice}} = 3.567 \times 10^{-5}$	J kg^{-1}
Latent heat for phase change between super-cooled water and ice (freezing and melting)	$h_{\text{vap-ice}} = 3.567 \times 10^{-5}$	J kg^{-1}
Latent heat for mass transfer between non-frozen membrane water and vapor	$h_{\text{nm-vap}} = 2.308 \times 10^{-6}$	J kg^{-1}
Entropy change in reaction	$\Delta S = -163, 110$	$\text{J kmol}^{-1} \text{K}^{-1}$

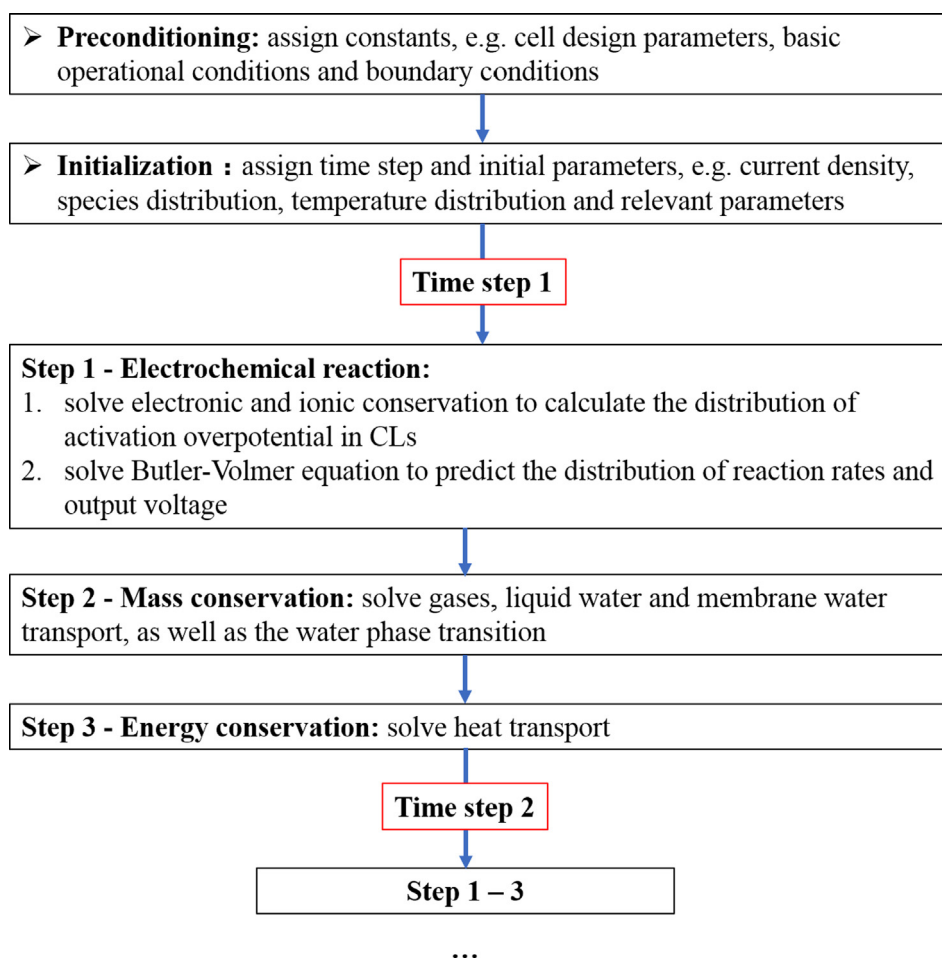


Fig. 3. Flow chart of the modeling calculation.

on the local reaction site inside CL, as displayed in Fig. 1. In this study, vapor, super-cooled water and membrane water generations are all considered. Water phase transition mechanisms in CL have been rebuilt, as shown in Fig. 2. Phase transition rates have been employed to quantify the phase change procedure in this work. Since there is still no general agreement on the issue that whether the membrane water and vapor could reach over-saturated state or not, both non-equilibrium and equilibrium phase transition models have been taken into account for assisting exploring the close-to-reality water transport behavior. Note that the membrane water and vapor could reach over-saturated in the non-equilibrium model, while for the equilibrium model the over-saturated state could not be reached. The goal is to provide a comprehensive baseline of the reasonable water production, transfer and phase transition mechanisms incorporated with the cold start operation of PEM fuel cell.

3.1. Membrane water generation

3.1.1. Non-equilibrium phase transition model

Considering that the catalyst particles are often wrapped inside the solid electrolyte cluster in the CL, the water generated in cathode semi-reaction is probably most membrane water. In this section, the product water is therefore assumed to be membrane water presenting inside the ionomer, aiming to discuss the resultant water transfer and cold startup behavior of PEM fuel cell. Wu et al. also proposed that the protons regularly migrate in the ionomer phase which results in the water production inside the electrolyte [36]. The membrane water generation is preliminarily considered to be reasonable but still needs to be further explored and validated through numerical and experimental data. The

water transfer behavior basically follows the non-equilibrium mechanism corresponding to the membrane water generation and phase transition illustrated in Fig. 2.

Fig. 4(a) gives a well validated operational capability of PEM fuel cell based on the membrane water generation starting up from $-20\text{ }^{\circ}\text{C}$ with constant current density at 100 mA cm^{-2} . The fuel cell design and operating parameters are consistent with the experimental test conducted by Tariji et al. [41], which suggests that the membrane water generation is reasonable to some extent. The initial water content is set as 3.4. This serves as an appropriate water content to maintain sufficient active area for the electrochemical reaction. Note that all the phase transition rates and water transfer rates remain to be 1 s^{-1} for the base operating conditions, as listed in Table 2. The modeling predicted cell voltage and the operational survival time performs coherent with the experimental test. However, the evolution of the cell voltage degradation at the shut-down stage is not perfectly fitted, for which the over-saturated membrane water content is probably responsible. Scrupulously, the over-saturated membrane water may result in the cell voltage plummeting when the cold startup failed, which does not smoothly match experimental performance decreasing, as displayed in Fig. 4(a). Owing to the hostilely low startup temperature, complete ice blockage of oxygen supply should be regarded as the main reason for the startup failure. The variation of membrane water content in cathode CL is demonstrated in Fig. 4(b). Note that saturated membrane water content serves as a function of the local temperature of membrane, as declared in Eq. (13). The crossing points with the saturated level of membrane water content refer to the beginning of membrane water desorption into liquid water and vapor. The difference in water content between the actual value and the saturated value reflects the driving

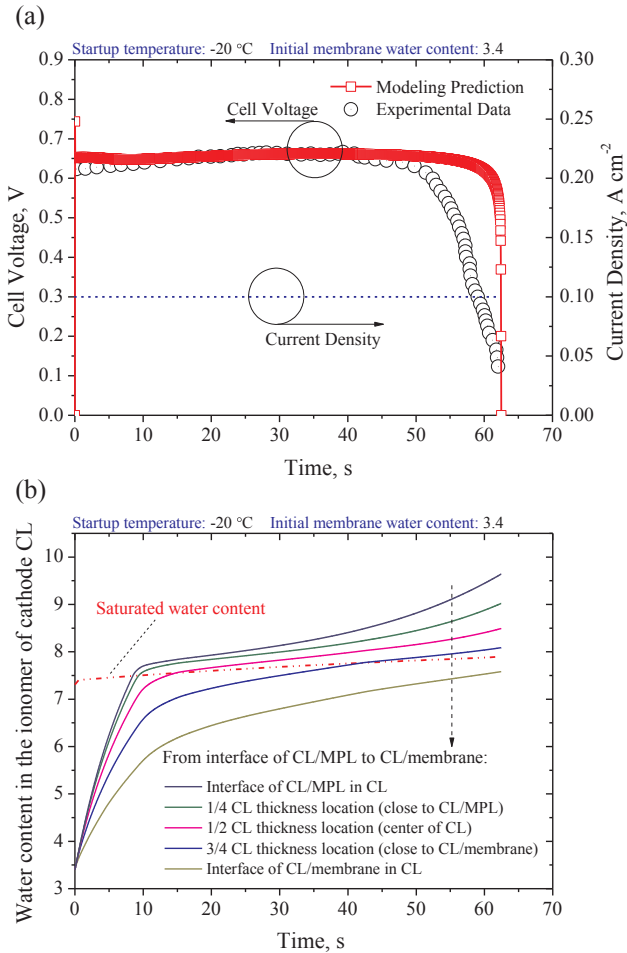


Fig. 4. (a) Comparison of cell voltage evolution between the present modeling prediction and experimental data [41]; and (b) evolution of water content in the ionomer of cathode CL, starting up from $-20\text{ }^{\circ}\text{C}$ with constant current density at 100 mA cm^{-2} and initial membrane water content of 3.4.

force for the membrane water desorption. The high level of membrane water content gives rise to the overestimated and more stable cell voltage at the end of the fuel cell operation compared with the practical experimental test. Furthermore, comparing to the liquid water and membrane water, the molar concentration of water vapor is theoretically much lower, leading to the weaker effects of desublimation, condensation and membrane absorption on the water transport behavior, as well as the evolution of fuel cell output. Liquid transfer behavior is observed to play a leading role in the water transition and transfer process. Meanwhile, in consideration of the less vapor availability, the ice is found mainly formed from liquid water. Therefore, in order to match the experimental data, the phase transition rates associated with liquid water is utilized as the main factors for the experimental verification, especially the water desorption rate ($\kappa_{\text{nf-m-lq}}$ between membrane water and liquid water).

With water desorption rate from membrane water to liquid water set as 1 s^{-1} , another set of comparison of modeling prediction and experimental data has been carried out in Fig. 5(a) [41]. Cell performance reaches a good agreement with the experimental data in both the cell voltage in the first half of operation and shut-down time, while the similar deviation of modeling predicted performance close to the end of the startup operation is also observed. The evolution of averaged membrane water content has been sketched in Fig. 5(b). It is seen that the averaged membrane water content achieves saturated at around 45–55 s after startup, and the performance deviation starts to expand from then on. This further implies that the possible reason for the

performance deviation comes down to the oversaturated membrane water and undervalued liquid water amount at the end of the operation. This further contributes to the ice accumulation slowing down as shown in Fig. 5(c), better ionic conductivity of polymer membrane, and thus overestimated cell output performance. The performance comparison at $-30\text{ }^{\circ}\text{C}$ with constant current startup at 50 mA cm^{-2} depicted in Fig. 6 further confirms the reliability of the membrane water generation and the general issue of the non-equilibrium model [42].

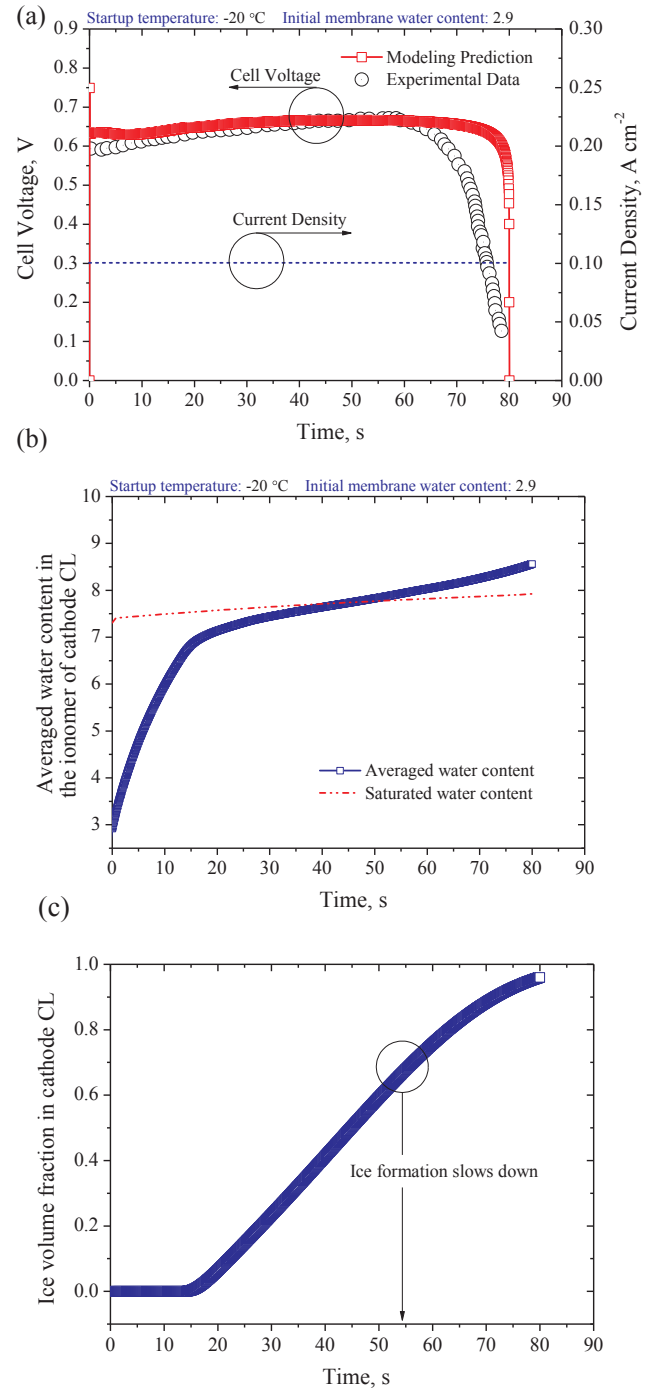


Fig. 5. (a) Comparison of cell voltage evolution between the present modeling prediction and experimental data [41]; (b) evolution of averaged water content in the ionomer of cathode CL; and (c) evolution of ice volume fraction inside cathode CL, starting up from $-20\text{ }^{\circ}\text{C}$ with constant current density at 100 mA cm^{-2} and initial membrane water content of 2.9.

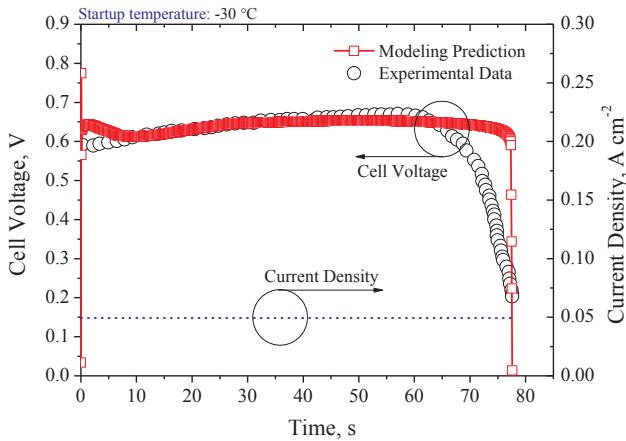


Fig. 6. Comparison of cell voltage evolution between the present modeling prediction and experimental data starting up from $-30\text{ }^{\circ}\text{C}$ with constant current density at 50 mA cm^{-2} [42].

3.1.2. Equilibrium phase transition model

In this section, the equilibrium phase transition model has been implemented, as illustrated by the simplified approaches for water generation and phase transition in Fig. 7. For the equilibrium model, several stages of water transfer behavior inside cathode CL should be addressed: (1) unsaturated water in ionomer and membrane water generation; (2) saturated water in ionomer and water vapor generation; (3) saturated water in ionomer, saturated vapor in pores and liquid water generation; (4) significant ice formation. It is noteworthy that both the membrane water and vapor could not get over-saturated in the equilibrium model. In other words, the equilibrium model does not explicitly solve the phase transition among the membrane water, liquid water and vapor. It is assumed that they are instantaneous balanced during the fuel cell operation.

The experimental comparison has been accordingly updated in Fig. 8(a). It is seen that the cell voltage decreases more smoothly and further approaches the experimental data during the deterioration stage

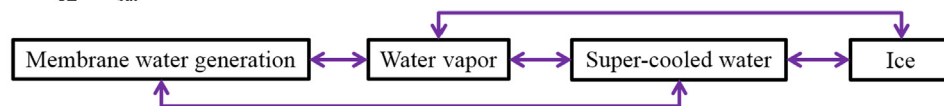
of the operation compared with the non-equilibrium model. It is worthwhile addressing that the water transfer behavior should be consistent with the non-equilibrium method before the membrane water attains saturated. However, according to the four stages mentioned above, liquid water generation is definitely accelerated by further water production. This could be also construed as the instantaneous membrane water desorption into liquid state, which is formulated to be liquid water production directly in the equilibrium model. Obviously, it also slightly decreases the modeling predicted membrane water content and the output performance on account of the negative role of liquid water generation playing in the reactant gases supply.

In detail, equilibrium model exhibits a much consistent behavior with the experimental data with respect to the voltage decrease, but only in the initial decrease, and does not accurately predict the shut-down time. In fact, the equilibrium phase transition cannot be taken as the real phase transition process. It should be viewed as an simplified equilibrium method for modeling the mass transfer of water. In this work, the phase change and mass transfer rates are employed to quantify the water phase transition in the non-equilibrium model. This makes it available approach the equilibrium state and experimental data through regulating the water transition processes. Basically, considering the negligible influences of water vapor on the startup operation, the most important factor comes from the water transfer between membrane water and liquid water, $\kappa_{\text{nm-lq}}$ (s^{-1}). In order to approach the equilibrium state of water transition, several membrane desorption (or absorption) rates have been representatively selected as 1, 2 and 3 s^{-1} . The objective of increasing the membrane water desorption rate is to accelerate membrane water desorption into liquid water and to promote the phase balance.

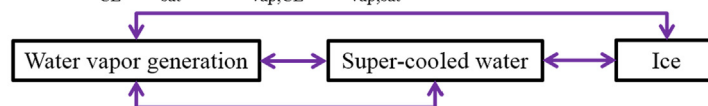
Through regulating $\kappa_{\text{nm-lq}}$, the startup performance has been updated based on the non-equilibrium model in Fig. 8(b). More reasonable evolution of cell output voltage has been achieved. It is evident that the membrane water desorption rate at around 3 s^{-1} is deemed to be very close to the water sorption equilibrium. The variation of membrane water content is given in Fig. 8(c). Increasing the membrane water desorption rate directly facilitates water transition from membrane

In cathode CL:

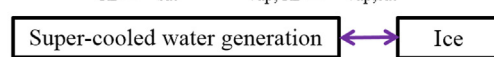
If $\lambda_{\text{CL}} < \lambda_{\text{sat}}$



else if $\lambda_{\text{CL}} \geq \lambda_{\text{sat}}$ and $C_{\text{vap,CL}} < C_{\text{vap,sat}}$



else if $\lambda_{\text{CL}} \geq \lambda_{\text{sat}}$ and $C_{\text{vap,CL}} \geq C_{\text{vap,sat}}$



In membrane:



In other domains:



Fig. 7. Equilibrium water generation and phase transition mechanisms.

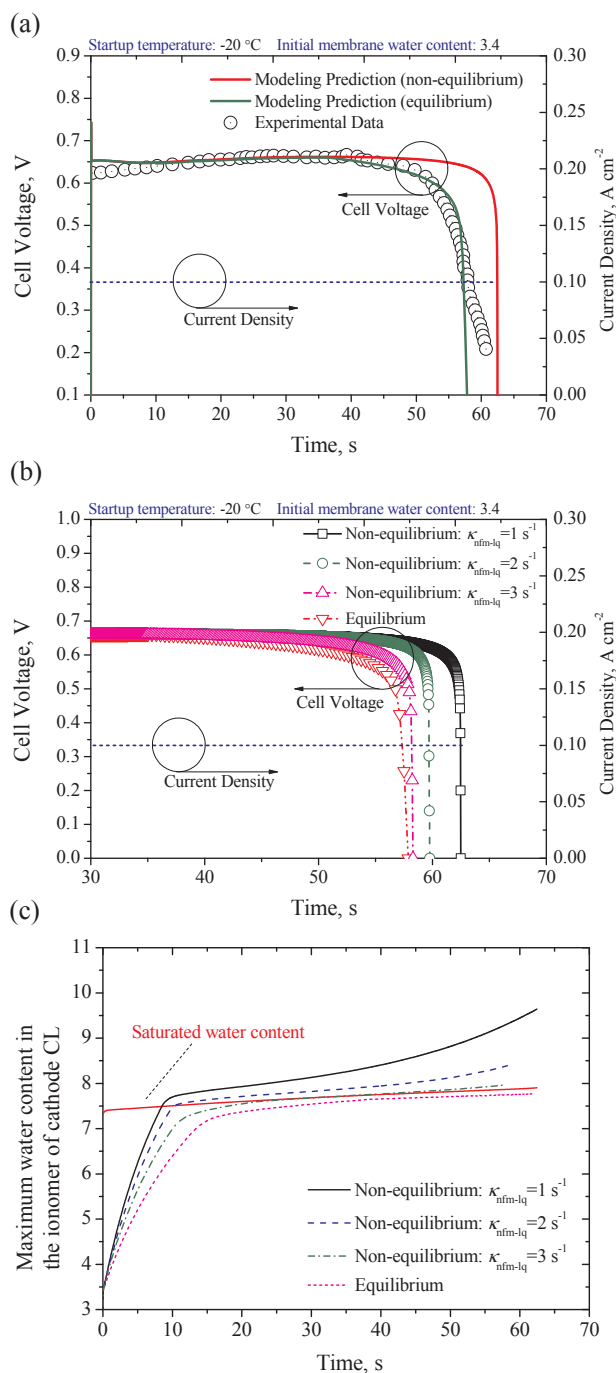


Fig. 8. Comparison of (a) cell voltage evolution between the present modeling prediction and the experimental data starting up from -20°C with constant current density at 100 mA cm^{-2} and initial membrane water content of 3.4 [41]; (b) cell performance between equilibrium and non-equilibrium phase transition; and (c) evolution of maximum membrane water content in cathode CL with various membrane water desorption rates ($\kappa_{nfm-lq}^* = 1, 2, 3 \text{ s}^{-1}$).

water to liquid, which is more prominent in the period before shutdown. It leads to not merely the smooth cell performance decreasing, but also the accurate survival operation time. It is worthwhile mentioning that the latent heat generation and consumption coupled with the water phase changes has been accounted in this study. However, owing to the weaker latent heat of phase transition on the order of 10^{-8} W m^{-3} compared to the prominent reversible and activation heat released in the reaction on the order of 10^{-6} – 10^{-7} W m^{-3} , phase transfer rates have insignificant effect on the thermal development.

Finally, the close-to-reality water transfer behavior inherent in the cold start of PEM fuel cell, especially inside the cathode CL, could be regularized into four stages: (1) membrane water generation and unsaturated water in ionomer; (2) water in ionomer reaching saturation; (3) over-saturated water in ionomer and quick desorption into liquid and vapor in pores; and (4) significant ice formation. This study is aiming to present basic framework of the reasonable cold start behavior, as well as the water generation, phase transition and transfer behavior inside the PEM fuel cell. It is expected that this study could offer useful insights into the reasonable modeling formulations for PEM fuel cell.

3.2. Liquid water and vapor generation

On the other hand, as liquid water may be present in the pores of CL due to the water sorption and desorption from the electrolyte, protons may flow through the liquid water. It indicates that electrochemical reaction could also take place in the liquid water cluster. The byproduct water may be also in liquid state accordingly, as delineated in Fig. 1. Nevertheless, since the ionic conductivity of proton in liquid phase is apparently worse than that in the ionomer, membrane water production theoretically predominates in the fuel cell operation, particularly in the early stage of cold starting up. Nonetheless, performance comparison will be still conducted in this section based on three different water production mechanisms in the electrochemical reaction, e.g. liquid, vapor and membrane water production, to briefly examine and point out their connotation to the numerical results. To brilliantly outline and compare startup capability, the phase transition rates of water are remained to be 1 s^{-1} for all the three cases.

Fig. 9(a) displays the time evolutions of cell voltage based on the three states of water generation with constant current operation at 0.1 A cm^{-2} starting up from -20°C . Initially the averaged cell voltage in the three cases varies with a similar pace because of the low initial membrane water content (3.4) and its slow pulling up. For the case of membrane water generation, a few seconds of delay in liquid and ice formation is observed due to the unsaturated water in the ionomer. This manifests that water generated in the ionomer offers benefits for preventing water freezing, sustaining better startup capability of PEM fuel cell, which is consistent with the delay in the induce time of ice formation [25–27]. On the other hand, the results indicate that water produced in liquid and vapor phase significantly motivates the deterioration of startup capability through directly speeding up ice precipitation from liquid water. It is also confirmed by examination of super-cooled water and ice distribution at 9 s after startup, as performed in Fig. 9(b). The corresponding cell performance deviates from the experimental results mentioned previously. Moreover, for the liquid water and vapor generation cases, the modeling predicted liquid and ice immediately generates from the beginning of the startup operation. It further illustrates the irrationality of the assumptions associated with the liquid water and vapor generation not only in the cold start modeling formulation, but also in the general modeling development of PEM fuel cell.

With regard to the case of membrane water generation, the liquid water evolution inside the cathode CL follows the slow increasing and rapid decreasing trend during the cold start operation, which implies that liquid water should mostly, even completely freeze at the shutdown moment of the PEM fuel cell, as demonstrated in Fig. 9(b). In detail, the liquid water and ice are generated at around 9 s after startup. Meanwhile, by checking the later mass transfer from non-frozen membrane water to liquid water in Fig. 4(b), it can be concluded that the liquid water and ice should be firstly transited from the saturated water vapor. Moreover, water prefers to freeze at the interface of CL and MPL due to the sufficient reactant supply and fast reaction kinetics. However, according to the corrected Butler-Volmer Equation, more serious ice blockage here results in the reaction kinetics degradation and gradually hinders the liquid water removal. Hence the liquid water

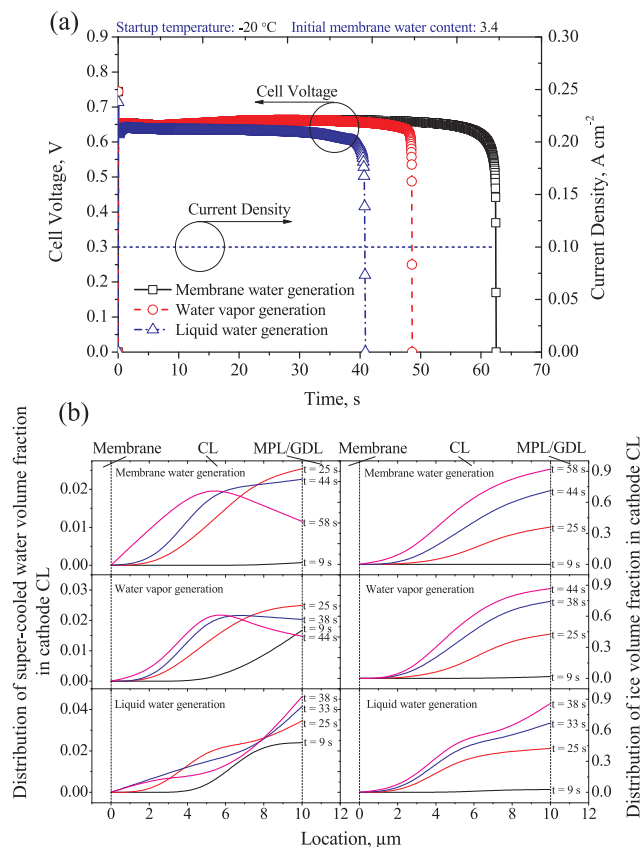


Fig. 9. (a) Startup performance and (b) distribution of super-cooled water and ice inside cathode CL based on different states of water generation: membrane water, liquid water and water vapor.

accumulation migrates from the interface of CL and MPL to the center of CL.

Therefore, it suggests that the modeling formulations for the water transfer inside the cathode CL follows the former-stated four stages with consideration of super-cooled water. In addition, it can be emphasized that water vapor production is almost impossible unless the PEM fuel cell is operating at a high temperature. Liquid water is possibly produced at a relatively low level in the electrochemical reaction. Hence this study suggests that the assumptions associated with the liquid water and vapor production should be cautiously used and even banished in the numerical works, especially for the dynamic modeling studies. Proper water transfer rates should be carefully chosen for the fuel cell operation under various cell design and operational conditions in order to achieve appropriate water transport behavior and reasonable modeling predicted results. From a physical standpoint, since the pore radius inside the electrolyte is always less than the order of 10 nm, the water flow prefers to be localized rather than connected within the narrow pores. Hence the water is often recognized to be membrane water in dissolved state in the ionomer instead of in liquid phase. Since the catalyst is usually assembled through spraying the catalyst-ionomer ink onto the polymer membrane, it reveals that the active area for the electrochemical reaction on the surface of catalyst particle should be mostly wrapped in the electrolyte, as shown in Fig. 1. As is well-known, the ionic group of sulfonic acid ($\text{SO}_3^- \text{H}^+$) will rarely retain less than three water molecules per proton. Subsequently, once a water molecule is formed at the triple-phase zone, it participates in the water molecules group connected with the sulfonic acid group. This further indicates that the membrane water production should play the leading role in the water generation mechanism in the electrochemical reaction.

4. Conclusions

In this study, an analytical multiphase model for the PEM fuel cell has been proposed to investigate the self-startup behavior from sub-freezing temperature. In consideration of the exiguity of the researches on the super-cooled water transfer characteristics in the PEM fuel cell at subfreezing temperature, the super-cooled water transport has been performed to enrich the water transfer mechanism in PEM fuel cell. In order to explore the reasonable water transfer behavior, water phase transition mechanisms have been rebuilt based on the states of water generation in cathode CL, including water vapor, membrane water and super-cooled water. The general water transfer behavior during cold start finally evolves into four stages: (1) membrane water generation and unsaturated water in ionomer; (2) water in ionomer reaching saturation; (3) over-saturated water in ionomer and quick desorption into liquid and vapor in pores; and (4) significant ice formation. The unreliable assumption associated with the vapor and liquid production in cathode reaction should be cautiously used, even prevented from the numerical model, especially for the dynamic modeling formulation.

The results indicate that the startup capability benefits a lot from water generation in membrane water state, which greatly helps delay the water freezing, and significantly slows down the ice growth inside the cathode CL. It is worthwhile addressing that the liquid water inside the cathode CL follows the slow increasing and rapid decreasing trend during the cold start operation, which indicates that liquid water should mostly, even completely freeze at the shut-down moment. On the other hand, liquid water accumulation gradually migrates from the interface of CL and MPL to the center of CL, impeded by the severe ice blockage and guided by the variation of reaction kinetics. Furthermore, the water transition associated with vapor is observed much weaker than any other water phase transitions owing to the intrinsic small amount of vapor in the pores. This study is aiming to present basic framework to possibly optimize the water balance inside the PEM fuel cell during cold start operation by reasonably regulating water generation, phase transition and transport. It is expected that this study could offer useful insights into the cold start enhancement of PEM fuel cell, as well as the reasonable numerical modeling formulations in the literature.

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